US INDEX CASE STUDY

Research Case Study

Subject: Linde plc - 2021 Russell US Indexes Reconstitution

Shareholder Structure

Q1. How frequently are changes to shares outstanding and free float implemented by Russell?

Shares outstanding and free float are reviewed quarterly:-

Third Friday of March, fourth Friday of June, third Friday of September, second Friday of December. For the June review, share and free-float updates are implemented regardless of size, every constituent's shares and float are re-sourced from filings and applied regardless of size. December refreshes market-cap ranking without a buffer, but float only updates if the change exceeds 3 percentage points. March and September are the most conservative reviews: shares-in-issue must move more than 1%, and float more than 3 percentage points (or 1 point if float is 15% or below).

Outside the quarterly cycle, qualifying primary or secondary offerings may trigger an update with at least two days' notice if they cause either a USD 1 billion investable-market-cap change, or both a 5% change in index shares and a USD 250 million investable-market-cap change.

Sources: Russell US Index Construction Methodology v7.0, Section 9.3.1–9.3.6, p.30.

Q2. Can you reconstruct the index float and share values given above for Linde? Show your working clearly and include any sources.

Linde's Q3 2020 10-Q (filed November 2020) gives 524,856,691 shares outstanding as of the cover date - this filing is used by Russell, not the later FY2020 10-K.

Holder	FTSE Russell Treatment
Vanguard — 42,775,562 shares, 8.16%	Portfolio Holding, restricted only at 30% — not restricted
BlackRock — 39,522,967 shares, 7.6%	Portfolio Holding, restricted only at 30% — not restricted
Directors & officers — 759,081 shares	Individual Investors, restricted at any size — restricted

Sources: Vanguard and BlackRock figures from Schedule 13G/A, Linde plc, event date 2020-12-31, EDGAR CIK 0001707925.

Shares outstanding (Q3 2020 10-Q) = 524,856,691
Restricted shares = 759,081

Public-Filing Free-Float Estimate

- Estimated free-float shares = Shares outstanding - Restricted shares
- Estimated free-float shares: 524,856,691 - 759,081 = 524,097,610
- Estimated free float: 524,097,610 / 524,856,691 x 100 = 99.8554%

The public proxy's issued insider shares produce approximately 99.9% when rounded conventionally to one decimal place. They do not, by themselves, reproduce Russell's stated 99.8%.

Reconciliation and Limitation

The 2021 proxy also disclosed 452,277 stock units for the director/officer group. Adding those units to the issued ordinary shares gives 1,211,358 share-linked interests and an implied float of 99.7692%, which displays as 99.8%. However, the stock units were not yet issued ordinary shares, and the supplied FTSE Russell methodology does not explicitly establish that they should be deducted from current free float. This is therefore a possible reconciliation, not a confirmed Russell calculation.

- Possible restricted share-linked interests: 759,081 + 452,277 = 1,211,358
- Possible implied free float: (524,856,691 - 1,211,358) / 524,856,691 x 100 = 99.7692%
- Index shares using Russell's published factor: 524,856,691 x 0.998 = 523,806,978

The 524,856,691 share count is reconstructed exactly. Public filings support a free-float estimate of 99.8554%; the displayed 99.8% can be reconciled only by including additional share-linked or restricted-holder data. Russell's precise calculation cannot be proven without its historical ownership input file.

Sources: Linde plc Form 10-Q, Q3 2020 (period ended 2020-09-30), cover page; Linde plc Schedule 14A (2021 proxy statement), beneficial ownership table; EDGAR CIK 0001707925.

Q3. What are the main SEC filings other than the Proxy (DEF 14A) that disclose shareholder information? Can information from those filings be used to forecast float changes for Russell?

Yes. SEC filings other than the DEF 14A can be used to forecast Russell free-float changes, but they are evidence inputs, not automatic index triggers. They must be public before the relevant Russell information cut-off and interpreted using FTSE Russell's holder-classification and implementation rules.

Filing	Discloses	Value	Notes
13D / 13G	Ownership above 5%	High	Tracks Portfolio Holding stakes against the 30% threshold
Forms 3/4/5	Insider ownership	Medium	Officer/director holdings restricted regardless of size
10-K / 10-Q	Shares outstanding	High	Primary source Russell uses
8-K	Offerings, buybacks, M&A	Medium	Signals a future change; timing set by Russell's buffers, not the filing date
S-1/S-3/424B	Offerings, lock-ups	High	Flags supply changes before they reach the index
13F	Manager positions, quarterly	Low	Filed by the manager not the issuer; lagged up to 45 days

Source: FTSE Russell Free Float Restrictions v3.4, pp.2-4.

The main SEC filings besides the DEF 14A are:

- Schedules 13D/13G and amendments: disclose major beneficial owners, generally above 5%, and their voting/dispositive powers.
- Forms 3, 4 and 5: disclose ownership and transactions by directors, officers and other insiders.
- 10-K and 10-Q: report shares outstanding, treasury shares, repurchases and issuances.
- 8-K and offering documents such as S-1, S-3 and 424B: disclose material capital events, primary/secondary offerings, selling shareholders and lock-ups.

Yes, these filings can forecast Russell float changes, but they are inputs rather than automatic triggers. Analysts must classify holders under FTSE Russell's free-float rules and apply the relevant information cut-offs, buffers and corporate-action timing.

For Linde, BlackRock's and Vanguard's 13G/A filings disclosed stakes of 7.6% and 8.16%. As portfolio holdings below the 30% restriction threshold, they would ordinarily remain in free float, while disclosed director/officer holdings from Forms 3/4/5 are restricted regardless of size. Linde's Q3 2020 10-Q also reported 524,856,691 shares outstanding, matching Russell's stated figure.

Sources: Russell US Index Construction Methodology v7.0, Section 9.3.9-9.3.12, p.31; FTSE Russell Free Float Restrictions v3.4, pp.3-4.

Country Classification

Q1. Describe the country classification methodology used by Russell.

Russell uses three Home Country Indicators - incorporation, headquarters, and most liquid exchange (by 2-year average daily dollar volume). If all three agree, that's the country, no further test needed. If they don't, Russell works through a 4-step test:

1. Assets test: assign the company to its primary asset location if it matches an HCI.
2. Revenue test: if assets are inconclusive, apply the same test using revenue.
3. Headquarters fallback: if both tests are inconclusive, assign the headquarters country.
4. BDI exception: if headquarters is in a benefit-driven incorporation country, assign the country of the most liquid exchange instead.

A few details that change the outcome in practice: regional data is only treated as inconclusive when a company also discloses more detailed country-level data for the same period - Russell prefers the more granular figure when both exist. If a company reports by region only, that data can still pass the test (the worked example below shows a region clearing the 20-point bar on its own). Negative asset or revenue figures are automatically inconclusive. Long-lived assets are preferred over total assets, and total revenue over net revenue, when more than one figure is reported.

Sources: Russell US Index Construction Methodology v7.0, Section 5.2-5.3, p.12; Section 5.5, p.13; Appendix B, pp.37-39.

Two of the methodology's own worked examples are useful here:

A company incorporated in Ireland, headquartered in Ireland, and trading in Ireland (alongside the US and Germany) resolves at Step 1 to Ireland - the test is whether the company is incorporated, headquartered, and traded in one common country, not whether its most liquid exchange happens to be elsewhere. The second example shows the 20-point gap test failing:

Highest country share	30% (the HCI country)
Next-highest country share	15%
Gap	30% minus 15% = 15 points -- below the 20-point bar, test fails
Result	Falls to Step 4 -- assigned by headquarters

Sources: Russell US Index Construction Methodology v7.0, Section 5.5, "Example 2: ABC company"; Appendix B, "Example by country", pp.13, 39.

Q2. What do you think the classification of Linde would have been at the 2020 and 2021 June Reconstitution.
Linde’s three Home Country Indicators were split in both year:

1. Incorporation: Ireland
2. Headquarters: United Kingdom
3. Most liquid exchange: United States, assuming NYSE exceeded Frankfurt by two-year average daily dollar trading volume.

Sources: Linde plc Form 10-K, FY2019 and FY2020 (filed 2021-03-01), cover pages, EDGAR CIK 0001707925.

The three HCIs don't agree - Ireland, UK, US - in either year, so Step 1 fails both times. But Step 2 doesn't automatically fail: Linde reports sales by region only (Americas, EMEA, APAC), with no country-level breakout, so the regional data is usable, not inconclusive. The test is whether the region containing the US HCI beats every other region by 20 points or more, using a two-year average of segment sales ending in the year before each rank day.

Period	Americas %	EMEA %	Difference	Result
2018-19 avg (2020 test)	46.5%	20.7%	25.8pp -- clears 20pp	US
2019-20 avg (2021 test)	38.7%	23.6%	15.1pp -- fails 20pp	falls to Step 4

Segment sales from Linde plc Forms 10-K, FY2018, FY2019, FY2020, EDGAR CIK 0001707925. Two-year average of each year's percentage share, per the methodology's own averaging method (Appendix B).

2020 clears the regional threshold on Americas, which contains the US HCI - US classification. 2021 doesn't clear it - the test falls through to Step 4, where headquarters (UK, non-BDI) decides the outcome.

2020 June Reconstitution	United States
2021 June Reconstitution	United Kingdom

Russell US Index Construction Methodology v7.0, Appendix B (BDI list excludes UK and Ireland), p.39.

This is the more defensible reconstruction: it produces an actual change between 2020 and 2021, consistent with Linde being removed from the Russell 1000 specifically at the 2021 reconstitution rather than at some earlier point.

Q3. If you have any uncertainty about your answer could you highlight where and suggest how you could increase your certainty?

The conclusion is well supported but provisional because four uncertainties remain:

1. Assets precede revenue: Russell tests geographic assets before revenue. The reconstruction uses regional sales because no conclusive geographic asset breakdown has yet been identified. Usable asset data would take priority and could alter the result.
2. Regional mapping: The analysis assigns the Americas region to the US HCI because the US is the only Linde HCI country in that region. This treatment should be confirmed against Russell's historical application of the regional-reporting rule.
3. Most-liquid exchange: NYSE is assumed to be the exchange HCI, but Russell uses two-year average daily dollar trading volume across all exchanges within each country. This should be calculated for NYSE and Frankfurt.
4. Historical rules and inputs: The supplied methodology is dated 2026, while the events occurred in 2020-2021. The historical methodology or Russell's internal geographic data may have differed.

To increase certainty, I would examine Linde's complete 2019 and 2020 10-K geographic notes for usable asset data; calculate two-year ADDTV for NYSE and Frankfurt; obtain the Russell methodologies effective at each reconstitution; and review Russell's historical country-assignment files or client notices.

Until those checks are completed, the most defensible conclusion is: Linde was likely classified as US in June 2020 and UK in June 2021, with moderate or high confidence rather than certainty.

Sources: Russell US Index Construction Methodology v7.0 (May 2026); Russell US Corporate Actions and Events Guide v6.8 (Oct 2025); FTSE Russell Free Float Restrictions v3.4 (eff. June 2025); Linde plc SEC filings (10-K FY2018, FY2019, FY2020 filed 2021-03-01; 10-Q Q3 2020; Schedule 13G/A -- Vanguard, BlackRock) via EDGAR CIK 0001707925.